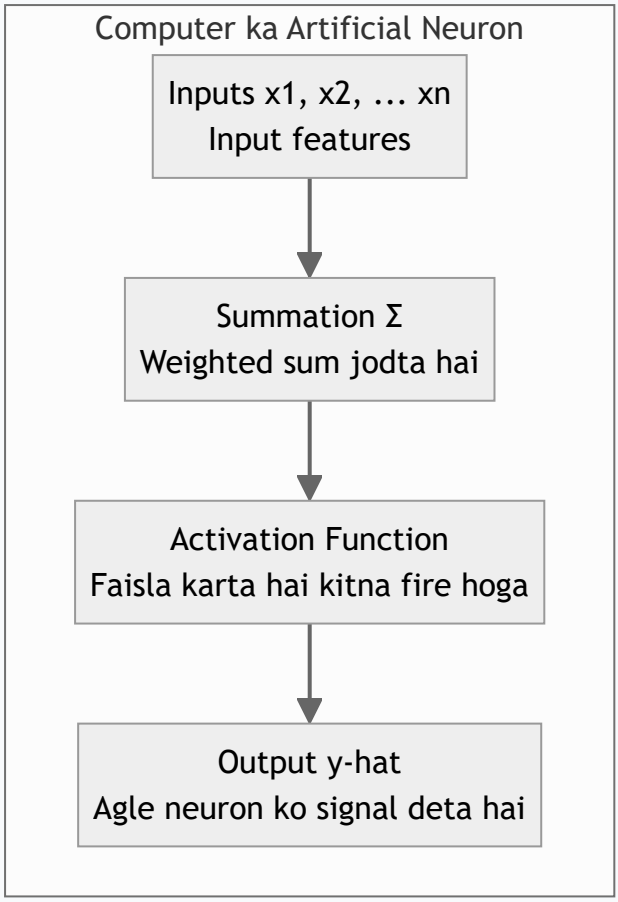
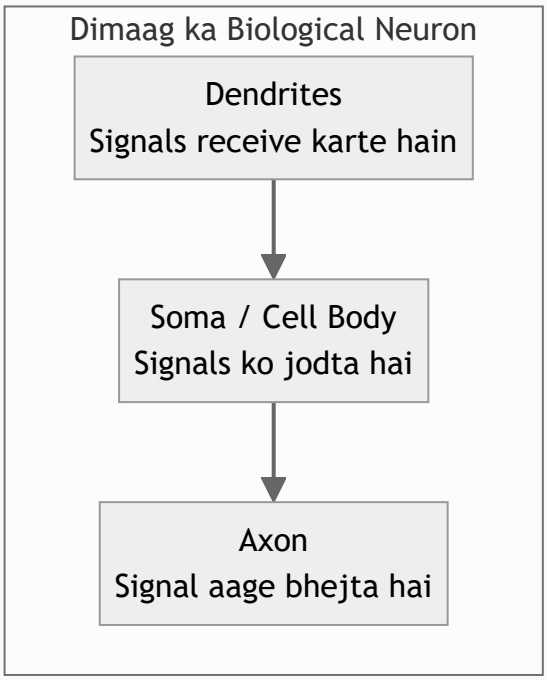
Submodule 36.03: Neural Networks kya hote hain? (Hinglish)

1. Insani Dimaag se Inspiration

Deep Learning ki shuruaat insani dimaag (human brain) ki working samajh kar hui. Hamare dimaag mein lagbhag **86 se 100 billion biological neurons** hote hain jo aapas mein synapses ke zariye jude hote hain.

Biological Neuron aur Artificial Neuron ka Comparison:


Computer ka Artificial Neuron
Inputs x1, x2, ... xn
Input features
Summation Σ
Weighted sum jodta hai
Activation Function
Faisla karta hai kitna fire hoga
Output y-hat
Agle neuron ko signal deta hai

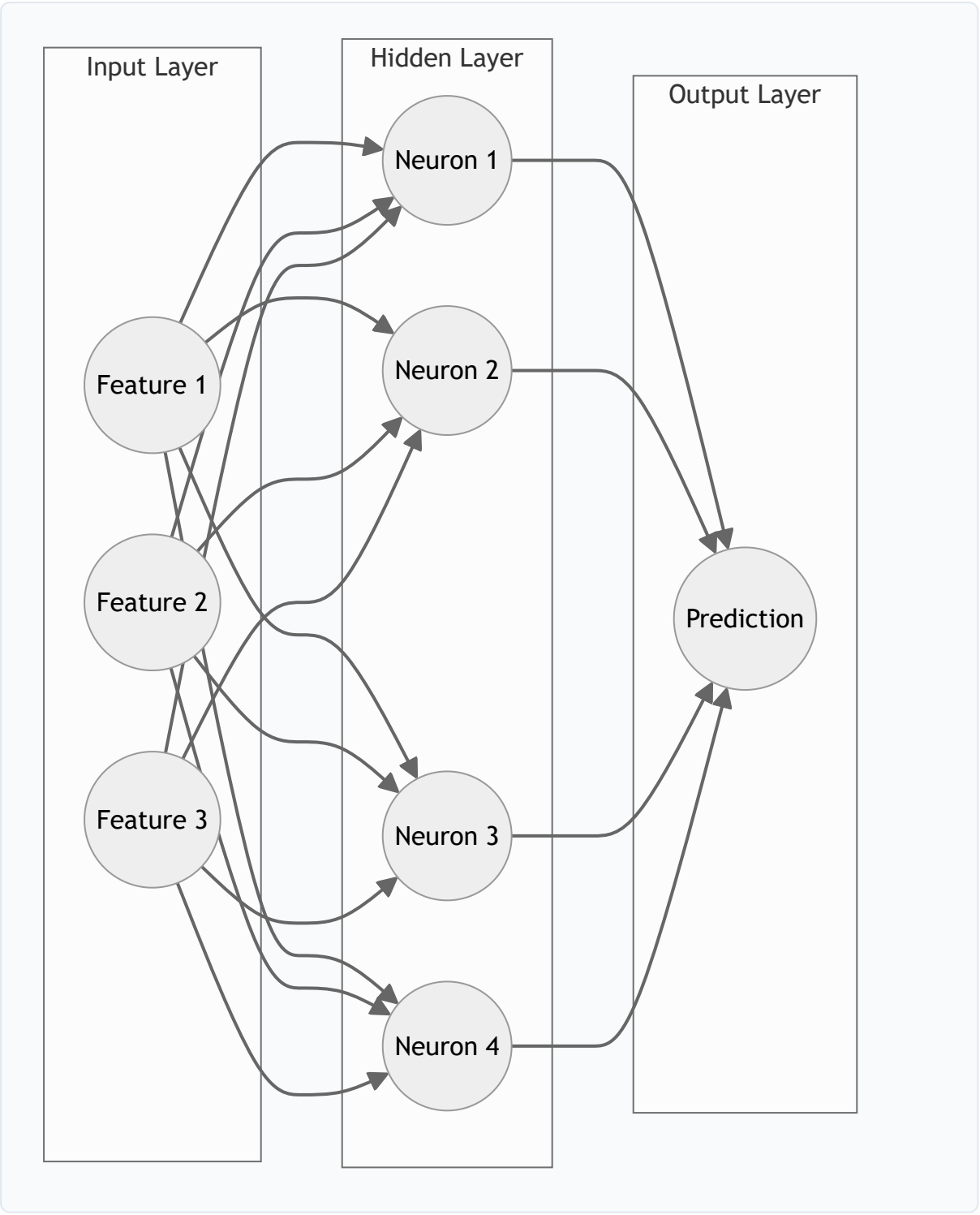

Dimaag ka Biological Neuron
Dendrites
Signals receive karte hain
Soma / Cell Body
Signals ko jodta hai
Axon
Signal aage bhejta hai

Biological Dimaag	Artificial Neural Network	Kaam kya hai?
Dendrites	**Inputs ($x_1, x_2, \ldots$)**	Bahar se information receive karna
Synaptic Strength	**Weights ($w_1, w_2, \ldots$)**	Har input ki importance/strength tay karna
Cell Body (Soma)	**Summation ($\sum w_i x_i + b$)**	Sabhi signals ka total jod nikalna
Firing Threshold	**Activation Function**	Faisla lena ki signal aage bhejna hai ya nahi
Axon	**Output ($\hat{y}$)**	Final answer aage pass karna

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2. Neural Network ka Structure (Layers)

Ek Artificial Neural Network teen types ki **Layers** se milkar banta hai:



1. **Input Layer**:
   - Yeh network ka gateway hai jahan data enter karta hai.
   - Har feature ke liye ek node hota hai (jaise house prediction ke liye `size`, `rooms`, `location` *o* 3 input nodes).
2. **Hidden Layer(s)**:
   - Input aur Output ke beech ki layers ko Hidden Layers kehte hain.
   - Yeh layers data ke andar ke complex patterns aur non-linear relationships seekhti hain.

- Jin networks mein 1 se zyada hidden layers hoti hain, unhe **Deep Neural Networks** kaha jata hai.

3. **Output Layer**:
   - Yeh final answer deti hai (jaise Spam hai ya nahi, ya phir car ki price kya hogi).